

NAME: \_\_\_\_\_ PERIOD: \_\_\_\_\_ DATE: \_\_\_\_\_

## Which Test Answers the Question Asked?

AP Statistics · Unit 3 · Topic 3.5 · B: 2026-12-08 · E: 2026-12-11

### [STAR] TODAY'S PROBLEM

A district of 2,400 students had bus ridership  $p_0 = 0.40$  before a new route. Before collecting data, the director asks whether the current proportion  $p$  has *increased*. An SRS of 180 current students finds 84 riders.

**Maya:** “Use  $H_0 : p = 0.40$  versus  $H_a : p > 0.40$  and a one-sample  $z$ -test. Check large counts with 72 and 108.”

**Theo:** “Use  $H_a : p \neq 0.40$  because samples can differ either way. Check large counts with the observed 84 and 96.”

#### Current SRS counts

| Group    | Bus | Other | Total |
|----------|-----|-------|-------|
| Students | 84  | 96    | 180   |

#### FIRST TAKE — before we discuss (5 min)

Does the director want to detect any change, or only more bus riders? Use a phrase from the study.

#### [BOOK] SET UP THE QUESTION BEFORE USING THE SAMPLE (keep on the page)

Define  $p$  as the **population proportion**. For a benchmark  $p_0$ , use  $H_0 : p = p_0$ . Choose  $H_a : p > p_0$ ,  $p < p_0$ , or  $p \neq p_0$  from the research question stated *before* seeing data;  $\hat{p}$  is sample evidence and never belongs in the hypotheses. For one categorical variable and one sample, use a one-sample  $z$ -test for a population proportion. Verify random data,  $n \leq 0.10N$  when sampling without replacement, and the null-expected counts  $np_0 \geq 10$  and  $n(1 - p_0) \geq 10$ .

Work (a)–(c) from the rules box:

**DOK 1** (a) Define the population parameter  $p$  in context and identify the null value  $p_0$ .

**DOK 2** (b) Compute  $\hat{p}$ , name the testing procedure, and verify the random, 10 percent, and large-counts conditions using the appropriate values.

**DOK 3** (c) In 3–4 sentences, choose a setup using the director’s word increased. Explain why the count check uses 72 and 108 rather than 84 and 96. State what different question Theo’s alternative would answer.

**Frame:** The warranted alternative is  $H_a : p$  \_\_\_\_\_ 0.40 because the director asked whether \_\_\_\_\_.

**Frame:** The large-counts check uses \_\_\_\_\_ and \_\_\_\_\_, not \_\_\_\_\_, because \_\_\_\_\_.

**EXIT — turn this in**

One thing the rules box changed about my first take: \_\_\_\_\_